

Erdős Problem 193 proof: notation and formula cheat sheet

Source of truth: paper/erdos193.tex, Sections 1–5. The proof ends before Section 6 (“Verification and evidence”).

1. Ambient conventions

Notation	Meaning
$\mathbb{N}$	Nonnegative integers $\{0, 1, 2, \dots\}$ in this paper.
$\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^3$	Integers, planar integer lattice, and spatial integer lattice.
$\mathbb{F}_2$	Field with two elements. Exponents reduced modulo 2 encode parity.
$ x $	Absolute value.
$\ u\ _1$	ℓ^1 norm; for $u = (u_x, u_y)$, $\ u\ _1 = u_x + u_y $.
$f \circ g$	Function composition with the right-hand map acting first: $(f \circ g)(z) = f(g(z))$.
$\nu_2(k)$	Two-adic valuation: the exponent of the largest power of 2 dividing nonzero integer k . The convention is $\nu_2(0) = \infty$.
q_j	Base-4 digit at place j ; $j = 0$ is the least significant digit.
x_j, y_j	Coordinate bits emitted at binary place j .
u_x, u_y	Coordinates of a planar vector u .

An infinite S -walk is a sequence $(P_a)_{a \geq 0} \subseteq \mathbb{Z}^3$ satisfying

$$P_{a+1} - P_a \in S \quad (a \in \mathbb{N}),$$

for one fixed finite step set $S \subseteq \mathbb{Z}^3$.

2. Hilbert decoder

State transformations

The decoder acts on a bit pair $(x, y) \in \{0, 1\}^2$ using

$$\begin{aligned} I(x, y) &= (x, y), & S(x, y) &= (y, x), \\ T(x, y) &= (1 - y, 1 - x), & C(x, y) &= (1 - x, 1 - y). \end{aligned}$$

Here I is the identity, S swaps the bits, T swaps and complements them, and C complements both. The state set is

$$K = \{I, S, T, C\} \cong \mathbb{F}_2^2,$$

with

$$S^2 = T^2 = C^2 = I, \quad S \circ T = T \circ S = C.$$

Thus every $g \in K$ is an involution: $g^{-1} = g$.

During a decode, $g \in K$ is a **mutable local variable** for the current orientation state. It starts at

$$g = I.$$

After the last digit q_0 has been processed, define

$$\sigma(w) = \text{the resulting value of } g.$$

Thus g denotes the changing intermediate state; $\sigma(w)$ denotes its final value. They are related, but they are not interchangeable during the scan.

Digit data

The four child positions, in U-shaped traversal order, are

$$c_0 = (0, 0), \quad c_1 = (0, 1), \quad c_2 = (1, 1), \quad c_3 = (1, 0).$$

The symbol r_q denotes the transition map selected by digit q ; it is not a numeric sequence. The lookup table is

$$r_0 = S, \quad r_1 = I, \quad r_2 = I, \quad r_3 = T.$$

In

$$g \longleftarrow g \circ r_q,$$

“update” means **replace the current orientation map g by the composed map $g \circ r_q$** after digit q emits its coordinate bits. The subscripted map $r_q \in K$ is unrelated to the scalar integer r used later in the collinearity proof.

For a length- ℓ base-4 word

$$w = q_{\ell-1} \cdots q_1 q_0,$$

the decoder reads digits from most significant to least significant, begins in state I , and at digit q_j performs

$$(x_j, y_j) = g(c_{q_j}), \quad g \longleftarrow g \circ r_{q_j}.$$

Reading order and output place run oppositely: $q_{\ell-1}$ is read first, but the pair emitted by q_j occupies binary place j .

The finite-order Hilbert point is

$$H_\ell(w) = \left(\sum_{j=0}^{\ell-1} x_j 2^j, \sum_{j=0}^{\ell-1} y_j 2^j \right).$$

The map H_ℓ is a bijection from length- ℓ base-4 words to $\{0, \dots, 2^\ell - 1\}^2$. Consecutive words represent planar lattice neighbors.

The finite-order induction also uses the convention that the order- ℓ route starts at $(0, 0)$ and ends at $(2^\ell - 1, 0)$.

Writing $N = 2^\ell$, bitwise action on an $N \times N$ child square is

$$S(x, y) = (y, x), \quad T(x, y) = (N - 1 - y, N - 1 - x).$$

The four transformed order- ℓ copies have start–end pairs

$$\begin{aligned} (0, 0) &\longrightarrow (0, N - 1), \\ (0, N) &\longrightarrow (N - 1, N), \\ (N, N) &\longrightarrow (2N - 1, N), \\ (2N - 1, N - 1) &\longrightarrow (2N - 1, 0), \end{aligned}$$

and junction displacements $(0, 1)$, $(1, 0)$, and $(0, -1)$. A suffix decoded from state g rather than I has state $g \circ h$ whenever the I -started suffix has state h , and its emitted route is the bitwise g -transform of the I -started route.

Padding, infinite path, and terminal state

Two leading zero digits do not change the point or terminal state:

$$H_{\ell+2}(00w) = H_\ell(w), \quad \sigma(00w) = \sigma(w).$$

For $n \in \mathbb{N}$, use any sufficiently long, even-length, zero-padded base-4 representation of n :

- $H(n) \in \mathbb{N}^2$ is its finite-order Hilbert point;

- $\sigma(n) \in K$ is the state left after decoding it.

The same symbol $\sigma(w)$ denotes the terminal state of a finite word. Context distinguishes word input from integer input.

The infinite path $H : \mathbb{N} \rightarrow \mathbb{N}^2$ is injective and has unit steps:

$$\|H(n+1) - H(n)\|_1 = 1.$$

Only digits 0 and 3 change the state. If $N_0(w)$ and $N_3(w)$ count those digits, then

$$\sigma(w) = S^{N_0(w) \bmod 2} T^{N_3(w) \bmod 2}.$$

Thus σ records exactly the parity pair $(N_0 \bmod 2, N_3 \bmod 2)$.

If digit q leaves outgoing state h , its incoming state and emitted pair can be recovered backward:

$$\begin{aligned} \text{incoming state} &= h \circ r_q, \\ (h \circ r_q)(c_q) &= h(r_q(c_q)) = h(c_q), \end{aligned}$$

because $r_q(c_q) = c_q$.

3. Two-adic chord invariant

For a nonzero planar chord $u = (u_x, u_y) \in \mathbb{Z}^2 \setminus \{0\}$, define

$$\begin{aligned} p(u) &= \min\{\nu_2(u_x), \nu_2(u_y)\}, \\ \epsilon(u) &= \begin{cases} 1, & \nu_2(u_x) = \nu_2(u_y), \\ 0, & \nu_2(u_x) \neq \nu_2(u_y), \end{cases} \end{aligned}$$

and

$$V(u) = 2p(u) + \epsilon(u).$$

Interpretation: $p(u)$ is the largest common binary scale of the two coordinates. The bit $\epsilon(u)$ says whether both normalized coordinates first become odd at that scale.

For every positive integer k ,

$$V(ku) = V(u) + 2\nu_2(k).$$

The factor 2 appears because each binary scale has two possible V -values: one coordinate first appears ($2j$), or both do ($2j+1$).

First-mismatch convention

For equal-length, even-padded words $q_{\ell-1} \cdots q_0$ and $q'_{\ell-1} \cdots q'_0$, define the first mismatch **from the low end** by

$$j = \min\{k \geq 0 : q_k \neq q'_k\}.$$

Digits $q_0, \dots, q_{j-1}$ form the common low suffix. If the two words have equal terminal state, their Hilbert coordinates agree below binary place j , and at place j :

- exactly one coordinate bit differs if $q_j - q'_j$ is odd;
- both coordinate bits differ if $q_j - q'_j \equiv 2 \pmod{4}$.

In the pair-law proof, the local names are

$$d = q_j, \quad e = q'_j,$$

and, for $m < n$,

$$n - m = 4^j M, \quad M \equiv e - d \pmod{4}.$$

A differing coordinate has the form

$$\pm 2^j + 2^{j+1}L = 2^j(\pm 1 + 2L),$$

where $L \in \mathbb{Z}$ and the parenthesized factor is odd.

Pair law

For distinct indices with the same terminal state,

$$\sigma(m) = \sigma(n) \implies V(H(n) - H(m)) = \nu_2(|n - m|).$$

This is the proof's central invariant.

4. Bounded-gap selector and lifted walk

For block index $a \geq 0$, let $g = \sigma(a)$ and choose a two-digit base-4 suffix through the offset map $\rho : K \rightarrow \{1, 3, 5, 13\}$:

Prefix state g	Chosen suffix	Decimal offset $\rho(g)$	Full state chain
I	11_4	5	$I \circ I \circ I = I$
S	01_4	1	$I \circ S \circ S = I$
T	31_4	13	$I \circ T \circ T = I$
C	03_4	3	$I \circ C \circ C = I$

Define the selected index

$$n_a = 16a + \rho(\sigma(a)).$$

Multiplication by $16 = 4^2$ appends the chosen two base-4 digits. Since every $g \in K$ is its own inverse,

$$\sigma(n_a) = I \circ g \circ g = I.$$

The selected gaps satisfy

$$\begin{aligned} n_{a+1} - n_a &= 16 + \rho(\sigma(a+1)) - \rho(\sigma(a)), \\ 4 &\leq n_{a+1} - n_a \leq 28. \end{aligned}$$

Two lift notations are used:

$$Q(n) = (H(n), n)$$

for the generic lift of an index, and

$$P_a = Q(n_a) = (H(n_a), n_a) \in \mathbb{Z}^3$$

for the selected walk. Every P_a has terminal state I through its index n_a .

5. Collinearity contradiction

The same-state lift theorem says: if $E \subseteq \mathbb{N}$ and σ is constant on E , then

$$\{Q(n) : n \in E\}$$

contains no three collinear points.

For a hypothetical collinear triple with $a < b < c$ in E , define adjacent index gaps and planar chords by

$$\begin{aligned} A &= b - a, & B &= c - b, \\ X &= H(b) - H(a), & Y &= H(c) - H(b). \end{aligned}$$

The corresponding spatial displacements are (X, A) and (Y, B) . Because $A, B > 0$, collinearity is equivalent to

$$BX = AY.$$

Factor the gaps as

$$\begin{aligned} A &= gr, & B &= gs, \\ g &= \gcd(A, B), & \gcd(r, s) &= 1. \end{aligned}$$

Cancelling g gives $sX = rY$. Coordinatewise divisibility yields an integer vector $W \in \mathbb{Z}^2 \setminus \{0\}$ with

$$X = rW, \quad Y = sW.$$

Applying the pair law and scaling law to the adjacent pairs gives

$$\begin{aligned} \nu_2(g) + \nu_2(r) &= V(X) = V(W) + 2\nu_2(r), \\ \nu_2(g) + \nu_2(s) &= V(Y) = V(W) + 2\nu_2(s). \end{aligned}$$

Hence

$$\nu_2(r) = \nu_2(s) = 0, \quad V(W) = \nu_2(g),$$

so coprime r and s are both odd.

For the endpoint pair,

$$c - a = g(r + s), \quad H(c) - H(a) = (r + s)W.$$

Set

$$t = \nu_2(r + s).$$

The pair and scaling laws give

$$\nu_2(g) + t = V((r + s)W) = V(W) + 2t = \nu_2(g) + 2t.$$

Therefore $t = 0$, which says $r + s$ is odd. But r and s are both odd, so $r + s$ is even: contradiction.

6. Finite step menu

Write one successive spatial displacement as

$$P_{a+1} - P_a = (d_x, d_y, d_z).$$

Its height increment is

$$d_z = n_{a+1} - n_a, \quad 4 \leq d_z \leq 28.$$

Unit Hilbert steps and the triangle inequality give

$$|d_x| + |d_y| = \|H(n_{a+1}) - H(n_a)\|_1 \leq n_{a+1} - n_a = d_z.$$

A fixed finite menu containing every realized step is therefore

$$S = \{(d_x, d_y, d_z) \in \mathbb{Z}^3 : 4 \leq d_z \leq 28, |d_x| + |d_y| \leq d_z\}.$$

This S is a convenient finite superset, not the minimal realized menu.

7. Letter index

Global or recurring symbols

Symbol	Type	Role
S	transformation, then set	Decoder swap $S(x, y) = (y, x)$ in Sections 2–4; finite spatial step set in the theorem and Section 5. See the collision warning below.
P_a	$\mathbb{Z}^3$	a th vertex of the selected infinite walk.
H_ℓ	finite words $\rightarrow \mathbb{N}^2$	Finite-order Hilbert decoder.
H	$\mathbb{N} \rightarrow \mathbb{N}^2$	Nested infinite Hilbert path.
σ	words or $\mathbb{N} \rightarrow K$	Terminal value of the mutable decoder state g after the last digit.
K	four-element set/group	Decoder state space $\{I, S, T, C\}$.
I, S, T, C	maps on $\{0, 1\}^2$	Four decoder transformations.
c_q	$\{0, 1\}^2$	Child position associated with digit $q \in \{0, 1, 2, 3\}$.
r_q	K	Digit-indexed transition map; after digit q , replace the current state g by $g \circ r_q$. Not the scalar r in the triple argument.
w	base-4 word	Finite decoder input.
ℓ	nonnegative integer	Word length / finite Hilbert order.
q_j, q'_j	base-4 digits	Digits of two compared words at place j .
x_j, y_j	bits	Coordinate bits emitted at place j .
$N_0(w), N_3(w)$	nonnegative integers	Counts of digits 0 and 3 in w .
$u = (u_x, u_y)$	nonzero $\mathbb{Z}^2$ vector	Generic planar chord.
$p(u)$	nonnegative integer	Common two-adic coordinate scale.
$\epsilon(u)$	$\{0, 1\}$	Whether the coordinate valuations are equal.
$V(u)$	nonnegative integer	Two-adic planar chord invariant.
ρ	$K \rightarrow \{1, 3, 5, 13\}$	State-dependent selector offset.
n_a	nonnegative integer	Selected Hilbert index in block a .
$Q(n)$	$\mathbb{Z}^3$	Generic lift $(H(n), n)$.
E	subset of $\mathbb{N}$	Arbitrary same-terminal-state index set.
d_x, d_y, d_z	integers	Coordinates of one successive step of P .

Proof-local symbols

Symbol	Scope	Meaning
g, h	Decoder arguments	g is the mutable current state, initialized at I , whose final value is $\sigma(w)$; h is an outgoing or common state used in reverse decoding.

Symbol	Scope	Meaning
d, e	First-mismatch lemma	Unequal digits q_j, q'_j .
j	First-mismatch argument	Least digit place where two base-4 words differ.
M	Pair-law proof	Integer remaining after $n - m = 4^j M$.
L	Pair-law proof	Integer collecting higher binary-place contributions.
k	Scaling lemma	Positive integer scalar; in the mismatch definition, a dummy digit-place index.
a	Selector sections	Block number indexing n_a and P_a .
a, b, c	Triple-free theorem	Three ordered indices in E ; this reuses a with a different local role.
A, B	Triple-free theorem	Positive adjacent gaps $b - a$ and $c - b$.
X, Y	Triple-free theorem	Adjacent planar chords.
g	Triple-free theorem	$\gcd(A, B)$; this reuses g with a different local role.
r, s	Triple-free theorem	Coprime reduced gap factors: $A = gr, B = gs$.
W	Triple-free theorem	Common nonzero primitive-direction multiple with $X = rW, Y = sW$.
t	Triple-free theorem	$\nu_2(r + s)$.

8. Collision and reading warnings

1. **S is overloaded.** In the decoder, S is the swap transformation. In the theorem and final section, $S \subseteq \mathbb{Z}^3$ is the finite step menu. Type and section determine the meaning.
2. **g is local and overloaded.** In the decoder, $g \in K$ is the mutable current state: it starts at I , and its final value is $\sigma(w)$. In the collinearity proof, the unrelated integer g means $\gcd(A, B)$.
3. **r_q and r are different objects.** The subscripted $r_q \in K$ is the transition map selected by base-4 digit q . The unsubscripted $r \in \mathbb{N}$ is the coprime reduced factor in $A = gr$.
4. **a changes role.** It is the block/walk index in n_a, P_a , but one of three arbitrary ordered indices $a < b < c$ inside the same-state lift theorem.
5. **σ is not a coordinate.** It is the accumulated terminal transformation left by decoding.
6. **Most-significant-first reading, least-significant place labels.** The decoder reads $q_{\ell-1}, \dots, q_0$, while q_j emits at binary place j .
7. **"First mismatch" means from the low end.** It is the least j , not the first digit encountered by the forward decoder.
8. **$H(n)$ is planar; $Q(n)$ and P_a are spatial.** The third coordinate is the Hilbert index itself.
9. **The menu is not the realized set.** The displayed finite S contains all possible successive displacements allowed by the proved bounds; it need not be minimal.